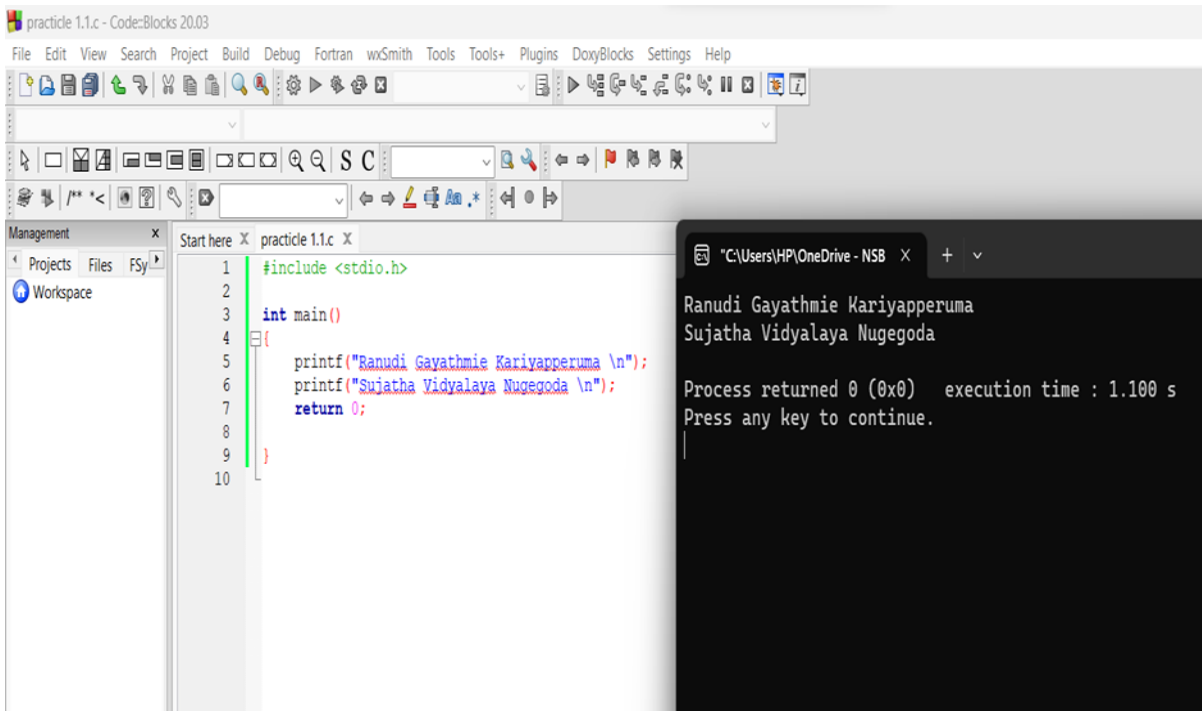


PRACTICAL 1

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1. `#include <stdio.h>`

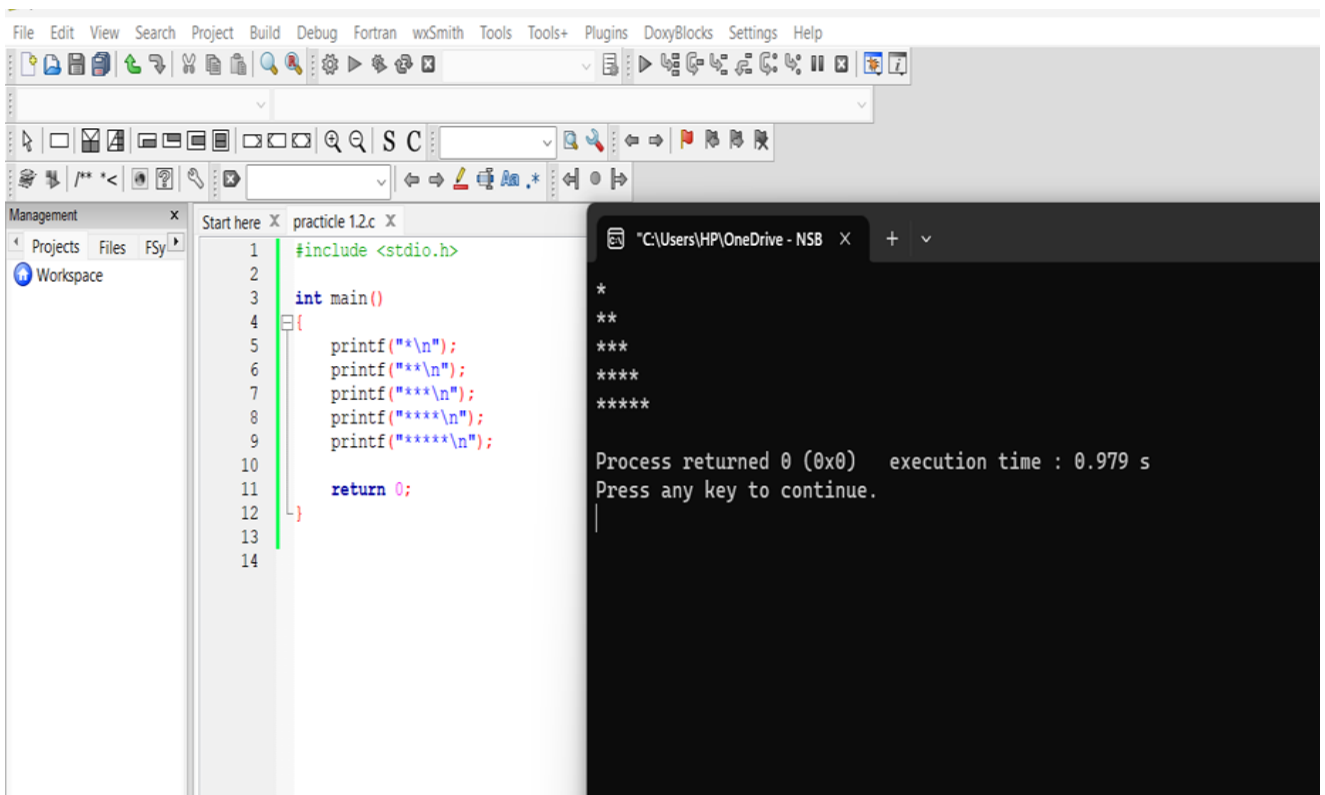
```
int main()
{
    printf("Ranudi Gayathmie Kariyapperuma \n");
    printf("Sujatha Vidyalaya Nugegoda \n");
    return 0;
}
```



```
2  #include <stdio.h>

int main()
{
    printf("*\n");
    printf("**\n");
    printf("***\n");
    printf("****\n");
    printf("*****\n");

    return 0;
}
```



The screenshot displays a C programming IDE with a menu bar (File, Edit, View, Search, Project, Build, Debug, Fortran, wxSmith, Tools, Tools+, Plugins, DoxyBlocks, Settings, Help) and a toolbar. The left sidebar shows a 'Management' pane with 'Projects', 'Files', and 'FSy' tabs, and a 'Workspace' section. The main editor window, titled 'practicle 1.2c', shows the following C code:

```
1  #include <stdio.h>
2
3  int main()
4  {
5      printf("*\n");
6      printf("**\n");
7      printf("***\n");
8      printf("****\n");
9      printf("*****\n");
10
11     return 0;
12 }
13
14
```

On the right, a terminal window titled '"C:\Users\HP\OneDrive - NSB"' shows the output of the program:

```
*
**
***
****
*****

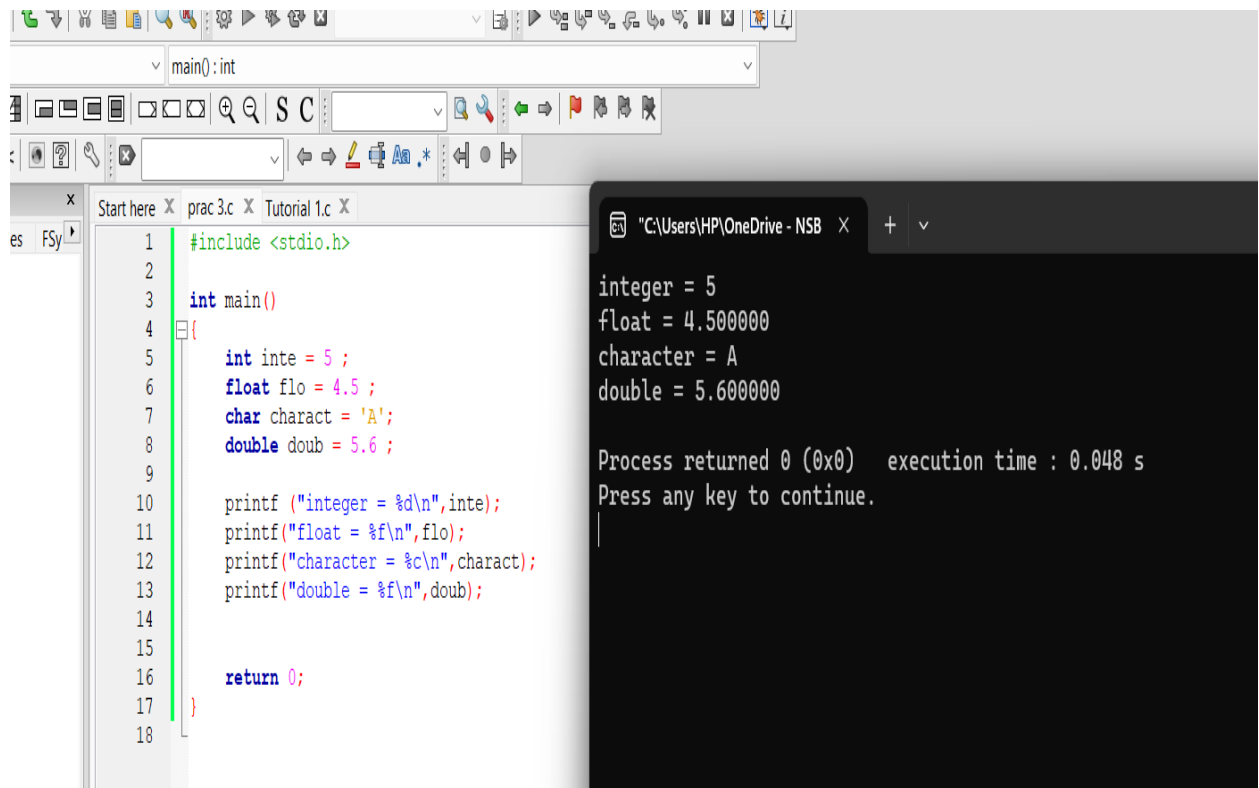
Process returned 0 (0x0)   execution time : 0.979 s
Press any key to continue.
```

3. #include <stdio.h>

```
int main()
{
    int inte = 5 ;
    float flo = 4.5 ;
    char charact = 'A';
    double doub = 5.6 ;

    printf ("integer = %d\n",inte);
    printf("float = %f\n",flo);
    printf("character = %c\n",charact);
    printf("double = %f\n",doub);

    return 0;
}
```



The screenshot shows a Windows environment with a code editor and a command prompt. The code editor displays the following C program:

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int inte = 5 ;
6     float flo = 4.5 ;
7     char charact = 'A';
8     double doub = 5.6 ;
9
10    printf ("integer = %d\n",inte);
11    printf("float = %f\n",flo);
12    printf("character = %c\n",charact);
13    printf("double = %f\n",doub);
14
15
16    return 0;
17 }
18
```

The command prompt window shows the output of the program:

```
integer = 5
float = 4.500000
character = A
double = 5.600000

Process returned 0 (0x0)   execution time : 0.048 s
Press any key to continue.
```

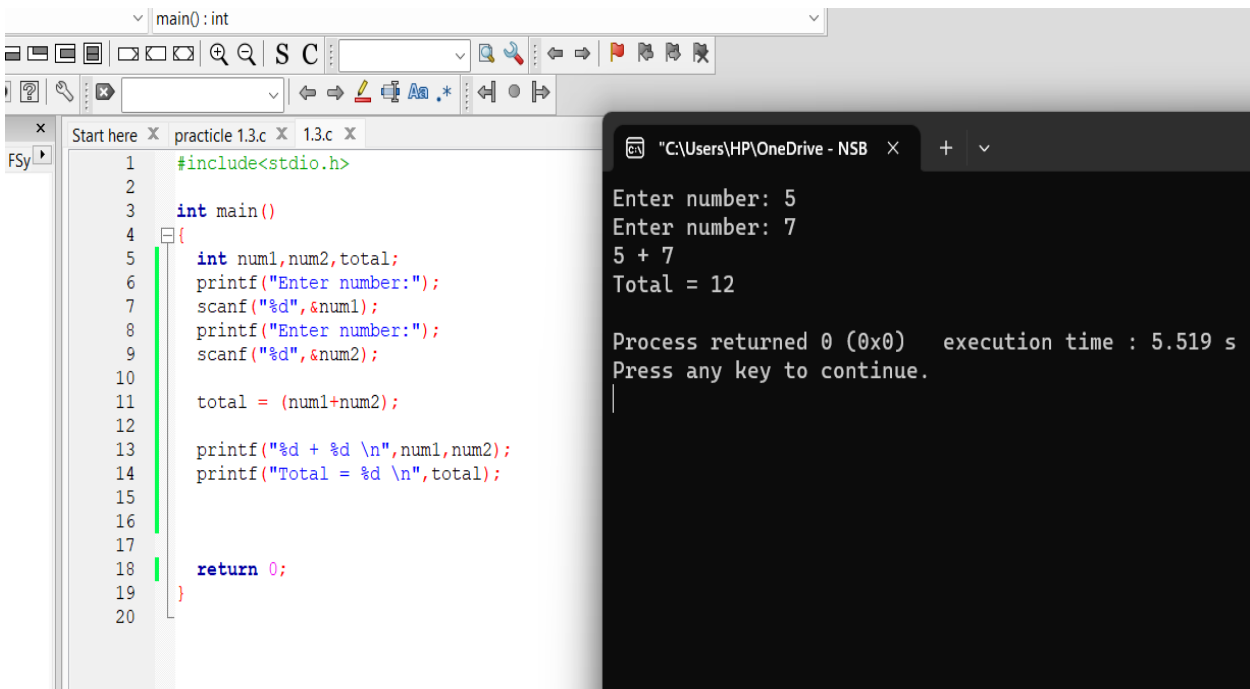
4. #include<stdio.h>

```
int main()
{
    int num1,num2,total;
    printf("Enter number:");
    scanf("%d",&num1);
    printf("Enter number:");
    scanf("%d",&num2);

    total = (num1+num2);

    printf("%d + %d \n",num1,num2);
    printf("Total = %d \n",total);

    return 0;
}
```



The screenshot shows a code editor window with a C program and a terminal window showing its execution. The code in the editor is as follows:

```
1 #include<stdio.h>
2
3 int main()
4 {
5     int num1,num2,total;
6     printf("Enter number:");
7     scanf("%d",&num1);
8     printf("Enter number:");
9     scanf("%d",&num2);
10
11     total = (num1+num2);
12
13     printf("%d + %d \n",num1,num2);
14     printf("Total = %d \n",total);
15
16
17
18     return 0;
19 }
20
```

The terminal window shows the output of the program:

```
Enter number: 5
Enter number: 7
5 + 7
Total = 12

Process returned 0 (0x0)   execution time : 5.519 s
Press any key to continue.
```

5 .#include<stdio.h>

```

int main()
{
    int num1,num2,Total;
    float Average;

    printf("Enter First Number:");
    scanf("%d",&num1);
    printf("Enter Second Number:");
    scanf("%d",&num2);
    Total = (num1+num2);
    printf("%d + %d \n",num1,num2);
    printf("Total = %d \n",Total);

    Average =(Total/2);
    printf("Average is %f ",Average);

    return 0;
}

```

The screenshot shows a C++ IDE with a file named 'practice 1.3.c'. The code in the editor matches the provided code block. The output window on the right shows the program's execution with the following text:

```

Enter First Number:20
Enter Second Number:30
20 + 30
Total = 50
Average is 25.000000
Process returned 0 (0x0)   execution time : 6.532 s
Press any key to continue.

```

```

6. #include<stdio.h>
   int main()
   {
       char Name[50];
       int birthyear,currentyear,Age;

       printf("Enter Your Name : ");
       scanf("%s",&Name);
       printf("Enter Your Birth Year : ");
       scanf("%d",&birthyear);
       printf("Enter Current Year :");
       scanf("%d",&currentyear);
       Age = (currentyear-birthyear);

       printf("Name is %s and Age is %d \n ",Name,Age);

       return 0;
   }

```

The screenshot shows a code editor window with a C program and a terminal window showing its execution. The code in the editor is as follows:

```

1  #include<stdio.h>
2  int main()
3  {
4      char Name[50];
5      int birthyear,currentyear,Age;
6
7      printf("Enter Your Name : ");
8      scanf("%s",&Name);
9      printf("Enter Your Birth Year : ");
10     scanf("%d",&birthyear);
11     printf("Enter Current Year :");
12     scanf("%d",&currentyear);
13     Age = (currentyear-birthyear);
14
15     printf("Name is %s and Age is %d \n ",Name,Age);
16
17
18     return 0;
19 }
20

```

The terminal window shows the output of the program:

```

Enter Your Name : Ranudi
Enter Your Birth Year : 2004
Enter Current Year : 2023
Name is Ranudi and Age is 19

Process returned 0 (0x0)   execution time : 12.014 s
Press any key to continue.

```

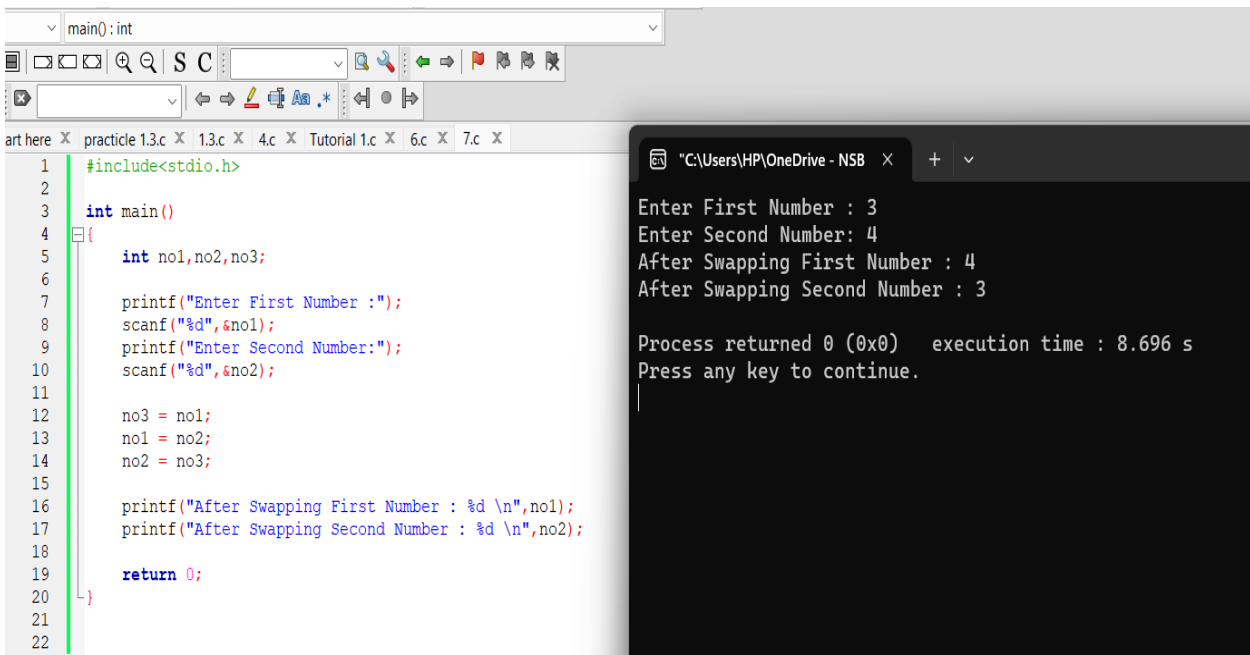
```
7.  #include<stdio.h>
    int main()
    {
        int no1,no2,no3;

        printf("Enter First Number :");
        scanf("%d",&no1);
        printf("Enter Second Number:");
        scanf("%d",&no2);

        no3 = no1;
        no1 = no2;
        no2 = no3;

        printf("After Swapping First Number : %d \n",no1);
        printf("After Swapping Second Number : %d \n",no2);

        return 0;
    }
```



The screenshot displays a C program being executed. On the left, a code editor window shows the source code for a program that swaps two numbers using a third variable. The code is as follows:

```
1  #include<stdio.h>
2
3  int main()
4  {
5      int no1,no2,no3;
6
7      printf("Enter First Number :");
8      scanf("%d",&no1);
9      printf("Enter Second Number:");
10     scanf("%d",&no2);
11
12     no3 = no1;
13     no1 = no2;
14     no2 = no3;
15
16     printf("After Swapping First Number : %d \n",no1);
17     printf("After Swapping Second Number : %d \n",no2);
18
19     return 0;
20 }
21
22
```

On the right, a console window shows the program's output. It prompts the user to enter two numbers, 3 and 4, and then displays the swapped values, 4 and 3. The console also shows the process returning 0 (0x0) with an execution time of 8.696 seconds, and a prompt to press any key to continue.

```
Enter First Number : 3
Enter Second Number: 4
After Swapping First Number : 4
After Swapping Second Number : 3

Process returned 0 (0x0)   execution time : 8.696 s
Press any key to continue.
```

```

8.  #include<stdio.h>
    main()
    {
        printf("The color: %s\n", "blue");// "%s" shown as "blue" value

        printf("First number: %d\n", 12345); // "%d" shown as "12345" value

        printf("Second number: %04d\n", 25);//Output shows Four values

        printf("Third number: %i\n", 1234);//Output shows the same value
        that entered

        printf("Float number: %3.2f\n", 3.14159);//This float number
        limits 2 decimals

        printf("Hexadecimal: %x\n", 255);//show the Hexadecimal value of
        255

        printf("Octal: %o\n", 255);//show the octal value

        printf("Unsigned value: %u\n", 150);//show the value same as the
        input

        printf("Just print the percentage sign %%\n", 10);// show the %
        mark
    }

```

The screenshot shows a C program being compiled and executed. The left pane displays the source code in a text editor, and the right pane shows the output of the program.

Source Code (Left Pane):

```

1  #include<stdio.h>
2  main()
3  {
4      printf("The color: %s\n", "blue");
5      printf("First number: %d\n", 12345);
6      printf("Second number: %04d\n", 25);
7      printf("Third number: %i\n", 1234);
8      printf("Float number: %3.2f\n", 3.14159);
9      printf("Hexadecimal: %x\n", 255);
10     printf("Octal: %o\n", 255);
11     printf("Unsigned value: %u\n", 150);
12     printf("Just print the percentage sign %%\n", 10);
13 }
14
15

```

Output (Right Pane):

```

The color: blue
First number: 12345
Second number: 0025
Third number: 1234
Float number: 3.14
Hexadecimal: ff
Octal: 377
Unsigned value: 150
Just print the percentage sign %

Process returned 0 (0x0)   execution time : 0.296 s
Press any key to continue.

```